

Managing Security Using SELinux



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Overview



Managing SELinux

- Security Enhanced Linux
- Managing Status
- Targeted Type Enforcement
- Changing Context
- Relocating User Home Directories





Understanding SELinux



SELinux



SELinux is a MAC, Mandatory Access Control, security system originally developed by the National Security Agency



Much of the Kernel module development has been run by both the NSA and Red Hat



Has been a part of Red Hat Enterprise Linux since version RHEL 4



Many services run with root access

SELinux controls access even to processes running as root



Understanding Security



**File system security
needs to be adjusted to
allow correct access to
directories**



**Firewalls need to be
adjusted to control
correct access to the
network**



**SELinux is a super-car
that used correctly will
protect your system, and
yes, it needs to be
adjusted**

SELinux Modes

SELinux has three operating modes:

1. **Enforcing:** Rules are enforced, and violations logged
2. **Permissive:** No rule enforcement in-place, violations are logged
3. **Disabled:** SELinux is not operational and no logging



Demo



Reading the current mode of SELinux

- with getenforce
- with sestatus
- directly from `/etc/selinux/config`

Changing the SELinux mode





Securing Mode Changes



```
$ sudo yum install -y policycoreutils setools setools-console setroubleshoot
```

Installing Tools

Adding additional tools will help you manage SELinux



```
$ getsebool -a  
$ sudo semanage boolean --list  
$ sudo setsebool secure_mode_policyload on (-P)
```

Preventing Runtime Changes to Mode

An administrator may quickly change the SELinux mode to allow something to happen that is not permitted

Setting the Boolean will require a reboot of the system to change the SELinux mode

Use the option -P to persist the change



Demo



Preventing runtime changes to the mode





SELinux Context



SELinux Type Enforcement

In the main, SELinux works with something called **type enforcement**. The SELinux type of a **source** must be compatible with the **target** SELinux type. If we need to customize content or if we make erroneous changes, operations may not work



```
$ ls -Z /etc/shadow
system_u:object_r:shadow_t:s0 /etc/shadow
$ ps -Z
```

LABEL	PID	TTY	TIME	CMD
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023	3751	pts/0	00:00:00	bash
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023	3784	pts/0	00:00:00	ps

List Contexts

In the main the targeted SELinux policy works with the context of processes, ports and files. The option **-Z** displays the context with most tools. Processes need to be authorized to access resources such as ports and files





SELinux Enforcement in Action



```
$ sudo -i
# ls -Z /etc/shadow
# chcon -t admin_home_t /etc/shadow
# chage -l vagrant
# ausearch -m avc -ts recent
# tail /var/log/messages
# sealert -l <alert-id>
# restorecon -v /etc/shadow
```

/etc/shadow

The /etc/shadow file is a critical system file. Processes can only access this if the correct context is set on the file and matches rules that allow the process access. Taking care not to break your system we can demonstrate SELinux at work.



Demo



Understanding Type Enforcement

- The wrong context





Tuning SELinux Contexts




```
$ man semanage-fcontext
$ sudo ls /etc/selinux/targeted/contexts/files
$ sudo mkdir /staff
$ sudo semanage fcontext -a -e /home /staff
$ sudo restorecon -v /staff
$ ls -ldZ /staff
$ sudo useradd -m -d /staff/u1 u1
$ sudo ls -ldZ /staff/u1
```

File Contexts

The file contexts used by restorecon are part of the current SELinux policy. Relocating user home directories, for example, we can create the top-level directory and store the definition by cloning the configuration of the existing home. This ensures the correct SELinux context on user home directories created below staff.



Demo



Understanding Type Enforcement

- Planning new home directories



Summary



SELinux is a Mandatory Access Control List

Modes

- Enforcing
- Permissive
- Disabled

Semanage Tools

SELinux Context

- -Z
- chcon / restorecon
- man semanage-fcontext

Up Next:

Congratulations

